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Influence of astrocyte-glioblastoma cell interactions on proliferation and migration

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Glioblastoma (GBM) is an aggressive tumor that proliferates and invades the brain tissue through complex interaction with the surrounding microenvironment. Of particular importance are the interactions between tumor cells and glial cells, amongst others astrocytes, which contribute to the modulation of tumor behavior.

This study compares the pro-tumorous features of two GBM cell lines (LN229 and U138) and astrocytes under monoculture and co-culture conditions. Since, inflammation is associated with tumor progression, the role of the pro-inflammatory activated astrocytes on proliferation and 3D spheroid generation was assessed.

Primary murine astrocytes and the murine astrocytic cell line C8-D1A were used under control condition, and in a pro-inflammatory A1-activated state. First, the cell-cell interactions were analyzed in 2D co-culture model. To characterize further cell-cell interactions, 3D co-cultures were established to mimic the tumor growth in a more complex microenvironment. Proliferation and cell cycle dynamics were characterized using FACS analysis and immunohistochemical staining (Ki67, BrdU), while molecules involved in cytoskeletal reorganization and cell-cell contacts in GBM cells were investigated by qRT-PCR and Western blot.

2D live cell-imaging revealed high migratory and proliferative profile of GBM cells, meanwhile human astrocytes exhibited a stable non-proliferative profile. Co-culture with astrocytes, significantly slowed down movement of the GBM cell lines. Analysis of cytoskeleton-associated mRNA in GBM cells demonstrated amongst others, elevation of RhoA and β -Catenin expression. In 3D co-culture systems, an initial acceleration of aggregation dynamics was present in both GBM cell lines. After 72 h of GBM-astrocyte co-cultures showed a reduced BrdU incorporation and an accumulation of cells in the G0/G1 phase, despite increased Ki67 expression.

Overall, these findings suggest that astrocytes mitigate cell growth and motility of the GBM cells contributing to the confinement of tumor spreading. Furthermore, pro-inflammatory activation of astrocytes seems to influence the proliferation and reduce growth behavior of GBM cells.



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